

Eric Xing

✉ e.xing@wustl.edu | 🎓 Scholar | 🌐 GitHub | 🌐 Website

RESEARCH STATEMENT

My research focuses on developing multimodal representation learning algorithms which serve impactful applications of visual search. I've developed state-of-the-art models for multimodal alignment [2, 6, 7], efficient multimodal retrieval [1, 13, 14], and multimodal reranking [1]. I publish in top computer vision and machine learning conferences, including CVPR, NeurIPS, ICCV, ECCV, and AAAI.

EDUCATION

Washington University in St. Louis

Ph.D. in Computer Science; Expected Graduation: Spring 2027

St. Louis, MO

Aug 2023 – Present

Western Kentucky University

B.S. in Computer Science with Minor in Mathematics, Summa Cum Laude; GPA: 4.00

Bowling Green, KY

May 2019 – Aug 2023

EXPERIENCE

Multimodal Vision Research Laboratory

Aug 2023 – Present

- Led collaboration across 4 universities with the National Center for Missing and Exploited Children to develop image search systems to support analyst investigations, resulting in multiple publications [1, 2, 13].
- Developed text-alignment algorithms to support the prediction of food system instability by NASA Harvest and the Famine Early Warning Systems Network.
- Led collaboration with WashU Public Health to develop text-guided surveying to automate the estimation of spatial inequities [9].
- Worked on three state-of-the-art algorithms for Earth representation based on spatial tessellation, spatio-temporal tessellation, and retrieval augmentation [8, 12, 15].
- Mentored BS/MS students across projects in representation learning, reinforcement learning, and applied collaborations.

Pinterest

Sep 2025 – Dec 2025

- Trained foundation models for multimodal search as an ML intern with the Advanced Technologies Group.
- Developed training, evaluation, and deployment infrastructure on production data.
- Proposed joint hypernetwork-encoder architecture for unified search and reranking across Pinterest's image indices, improving recall and reranking scalability.

Student Researcher @ Penn State

May 2022 – Aug 2023

- Proposed the novel authorship obfuscation system ALISON based on greedy replacement of important part-of-speech sequences [3].
- Demonstrated that ALISON achieves greater attack success while maintaining a higher degree of semantic preservation and significantly reducing runtime.
- Performed label entropy, parameter, and interpretability analyses on ALISON.

SELECTED PUBLICATIONS

- [1] **Xing, E.**, Stylianou, A., Pless, R., & Jacobs, N. *QuARI: Query Adaptive Retrieval Improvement*. The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**NeurIPS'25**). [link]
- [2] **Xing, E.**, Kolouju, P., Pless, R., Stylianou, A. & Jacobs, N. *ConText-CIR: Learning from Concepts in Text for Composed Image Retrieval*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025 (**CVPR'25**). [link]
- [3] **Xing, E.**, Venkatraman, S., Le, T., & Lee, D. *ALISON: Fast and Effective Stylometric Authorship Obfuscation*. The 38th Annual AAAI Conference on Artificial Intelligence (**AAAI'24**). [link]
- [4] **Xing, E.**, Liu, L., Xing, X., Qu, Y., Jacobs, N., & Liang, G. *Neural Network Decision-Making Criteria Consistency Analysis via Inputs Sensitivity*. 2022 International Conference on Pattern Recognition (**ICPR'22**), pp. 2328-2334. [link]

- [5] Kolouju, P., **Xing, E.**, Pless, R., Jacobs, N., & Stylianou, A. *good4cir: Generating Detailed Synthetic Captions for Composed Image Retrieval*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops 2025 (**CVPRW'25**). [link]
- [6] Khanal, S., **Xing, E.**, Dhakal, A., Sastry, S., Xiong, Z., Ahmad, A., & Jacobs, N. *PSM: Learning Probabilistic Embeddings for Multi-scale Zero-Shot Soundscape Mapping*. The 32nd ACM Multimedia Conference (**ACMMM'24**). [link]
- [7] Sastry, S., Dhakal, A., **Xing, E.**, Khanal, S. & Jacobs, N. *Global and Local Entailment Learning for Natural World Imagery*. The IEEE/CVF International Conference on Computer Vision 2025 (**ICCV'25**). [link]
- [8] Cher, D., Iqbal, H., **Xing, E.**, Wei, B., & Jacobs, N. *Tessellating The Earth*. European Conference on Computer Vision (Accepted to **ECCV'26**).
- [9] Leão, A., Banda, B., **Xing, E.**, Gudapati, S., Ahmad, A., Lin, J., Sastry, S., Jacobs, N. & Reis, R. *Applications of Artificial Intelligence in Public Health: Analyzing Built Environment and Addressing Spatial Inequities*. Journal of Public Health, 2025. [link]
- [10] Qiao, F., Xiong, Z., **Xing, E.**, & Jacobs, N. *Towards Open-World Generation of Stereo Images and Unsupervised Matching*. The IEEE/CVF International Conference on Computer Vision 2025 (**ICCV'25**). [link]
- [11] Wei, B., Sastry, S., Cher, D., **Xing, E.**, & Jacobs, N. *TerraDiT-Ω: Unified Spatial Control for Satellite Image Synthesis with Any Geospatial Primitive*. European Conference on Computer Vision (Accepted to **ECCV'26**).
- [12] Dhakal, A., Sastry, S., Khanal, S., Ahmad, A., **Xing, E.**, & Jacobs, N. *RANGE: Retrieval Augmented Neural Fields for Multi-Resolution Geo-Embeddings*. The IEEE/CVF Conference on Computer Vision and Pattern Recognition 2025 (**CVPR'25**). [link]

UNDER REVIEW

- [13] **Xing, E.**, Li, V., Cher, D., & Jacobs, N. *QuASI: Query-Adaptive Sparse Indexing for Multimodal Search*. Annual AAAI Conference on Artificial Intelligence (**AAAI'27** Submission).
- [14] **Xing, E.**, Wei, B., Cher, D., & Jacobs, N. *Learning to Allocate Tokens for Vision Transformers*. Conference on Neural Information Processing Systems (**NeurIPS'26** Submission).
- [15] Cher, D., **Xing, E.**, Li, K., Wei, B., Corley, I., & Jacobs, N. *Chronosphere: Space-Time Tessellation of Local Climate Experts*. Annual AAAI Conference on Artificial Intelligence (**AAAI'27** Submission).
- [16] Cher, D., Wei, B., Sastry, S., **Xing, E.**, Jarosz, P., Sandvig, R., & Jacobs, N. *Phenological Plant Embeddings*. Conference on Neural Information Processing Systems (**NeurIPS'26** Submission).

AWARDS AND SERVICE

McKelvey School of Engineering CSE Outstanding Scholar Award
 Honors Designation (top 15% of PhD students) – WashU Computer Science and Engineering
 Computing Research Association Outstanding Undergraduate Researcher – Honorable Mention
Reviewer: NeurIPS, CVPR, ICCV, ECCV, AAAI, WACV, SIGIR, ACMMM

ADDITIONAL

Languages: Python, C/C++, Java, JavaScript, SQL, Bash, MATLAB
Tools: Docker, Kubernetes, AWS, Git, Weights & Biases
Libraries: PyTorch, PyTorch Lightning, Ray, Optuna, HuggingFace, DeepSpeed, TensorFlow, NumPy, Pandas, Scikit-Learn, Stanza, NLTK, Captum, Matplotlib
 US Citizen